Eco-Risk Alpha

Measuring the Green Discount in Counterparty CVA

Summer School 2025 – Data Science for Sustainable Finance and Economics

Case Study

- **CVA and Counterparty Risk**

- **Use of CVA in real world**
  - Banks uses **CVA** to cover potential losses from counterparty default.

- **Empirical Evidence**
  - **Mixed results**: some find measurable ESG discount in the CVA, others argue sustainability is already priced.

Dataset Available

- ~60 companies
- CDS spreads 5y
- ESG Data:
  - ESG score
  - ESG pillars
  - Carbon Intensity
- Financial foundamentals:
  - Market Cap
  - Total Assets.
  - Debt-to-Equity
  - EBITDA Margin
  - Leverage Ratio
- Expected Positive Exposure for a 5-year payer IRS
- EUR Zero rates

A Link Between ESG and Credit Risk

Do firms with stronger ESG performance exhibit lower default risk and tighter credit spreads?

1. Prepare the dataset

2. Build statistical and ML models to explain CDS spreads

3. Translate spreads into CVA to estimate a green credit premium

Data Cleaning

Data Cleaning

- fix formatting
- remove the duplicates
- handle missing values

sector-wise median

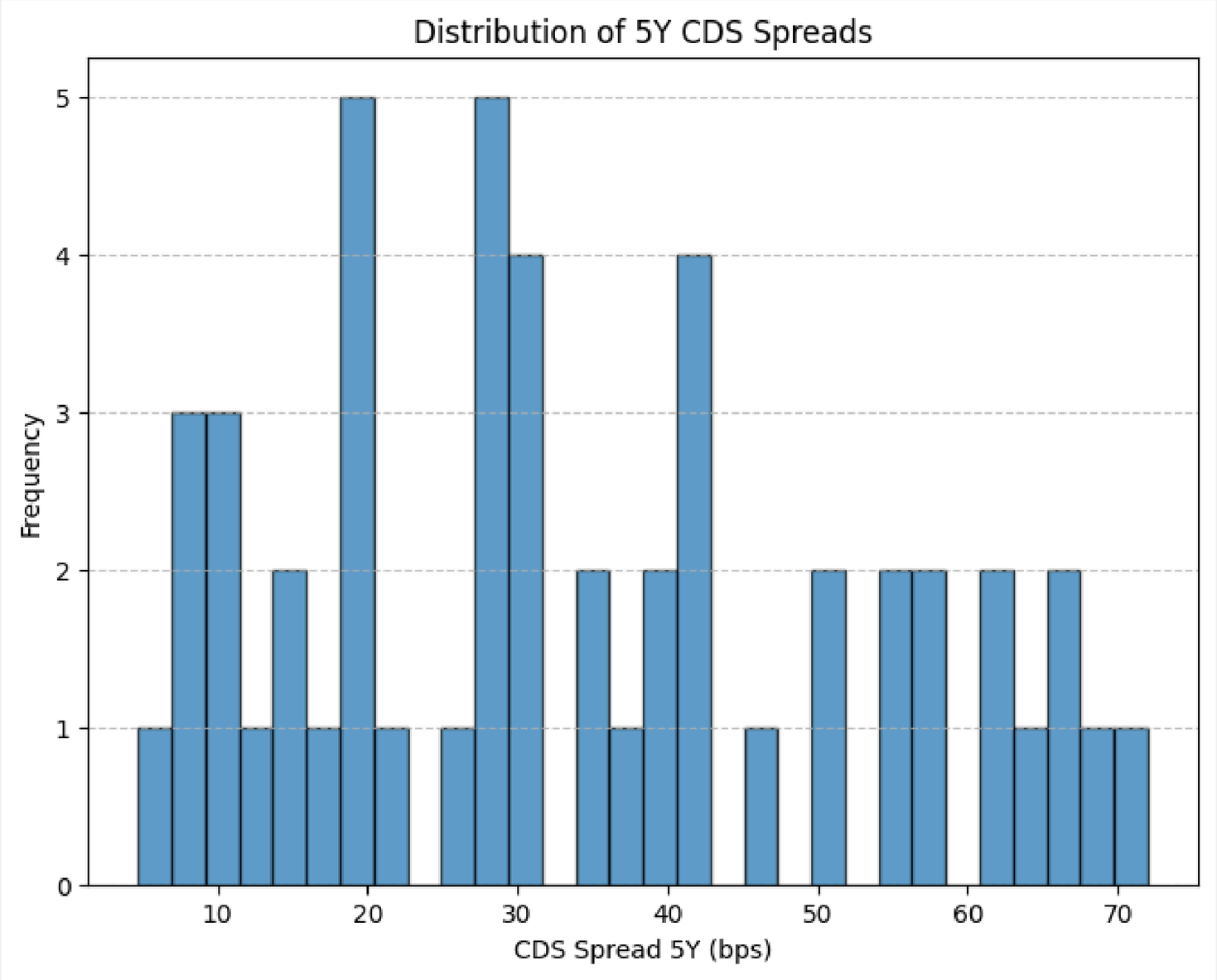
$$ESG = 0.5 \cdot E + 0.3 \cdot S + 0.2 \cdot G$$

Dealing with outliers

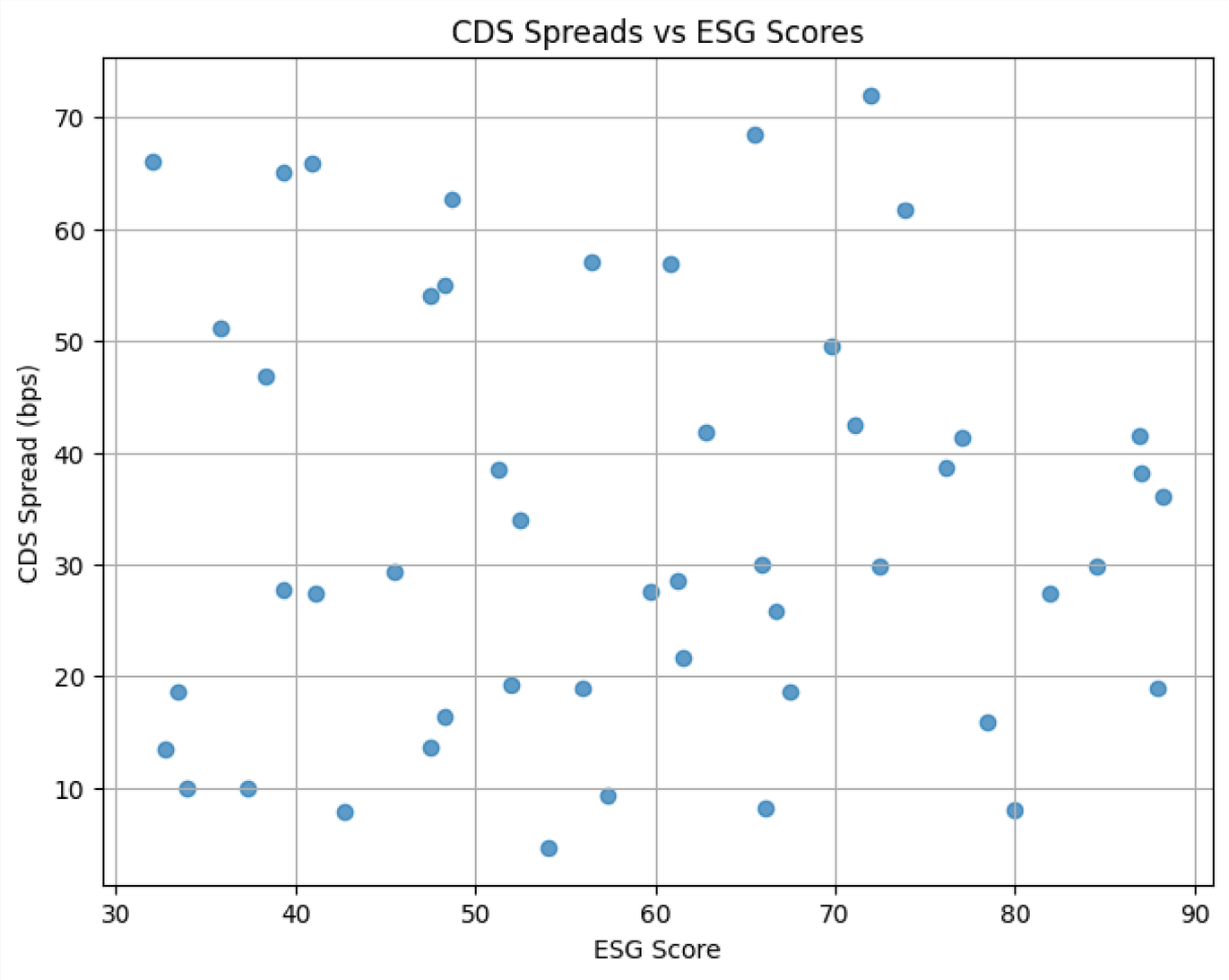
- Boxplot
  - Carbon Intensity
  - Market Cap
- are the outliers real or are they errors?
  - Log-transform

Visualization

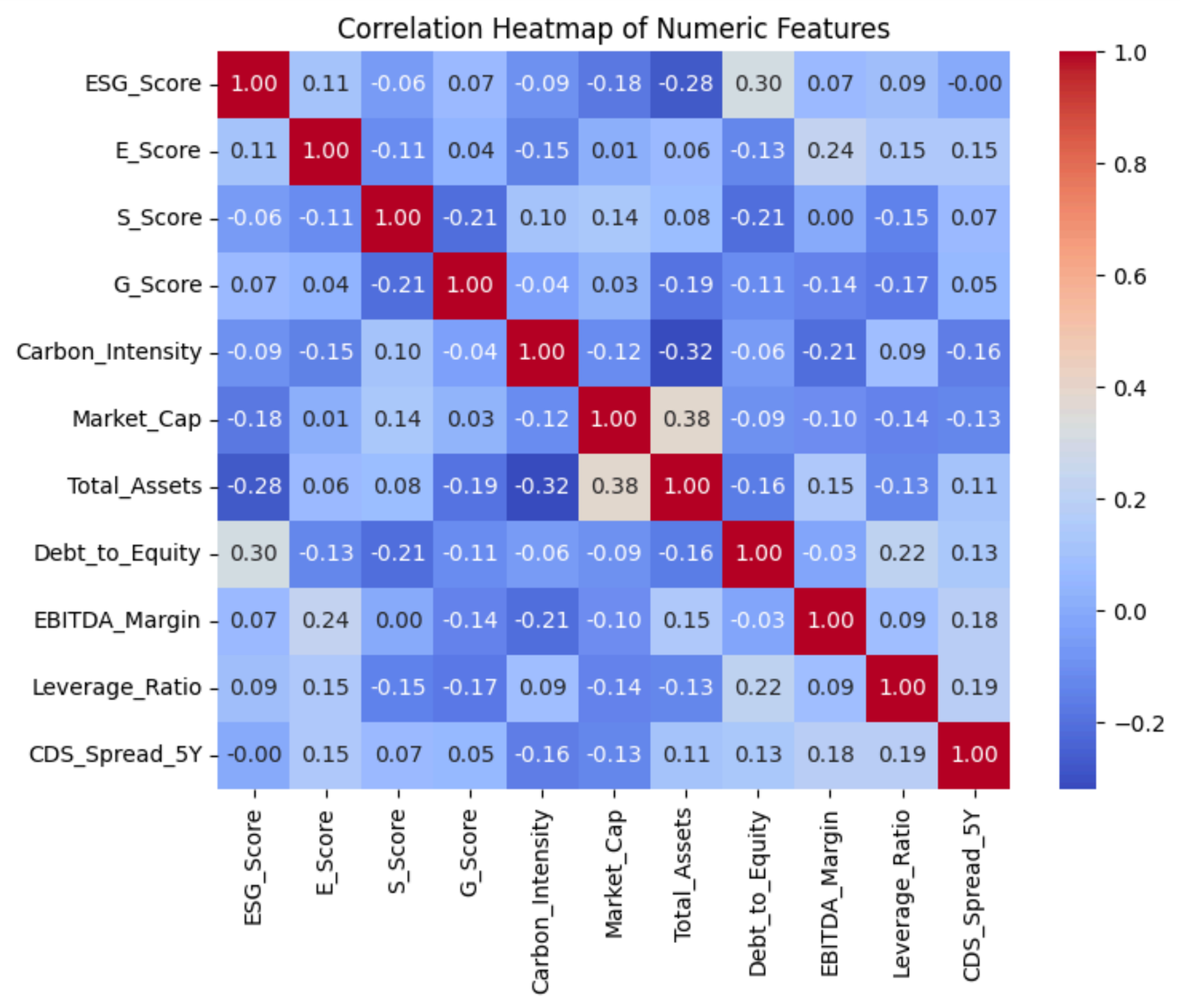
The Target Variable: CDS_Spread_5Y



No clear relation between CDS spreads and ESG scores



Correlation heatmap



Disclaimer

Only 50 entries

Numerical findings are not reliable!

Covariate Selection

How To Proceed?

Theoretical observations

+

Optimization of numerical performance

- Total_Assets and Market_Cap are both size measures
- ESG_Score and Carbon_Intensity both reflect environmental performance
- Leverage_Ratio and Debt_to_Equity are both measures of financial leverage

The models for prediction

- Linear regression
- Random forest
- Gradient boosting

Linear Regression

Ordinary Least Squares
+
White’s HC0 heteroskedasticity-robust SEs

Covariate selection

- ESG_Score
- Total_Assets
- Leverage_Ratio
- EBITDA_Margin

Criticalities

- Not enough data to support complex models
- Very weak explanatory power
- Regularization fails: shrinks everything to zero
- Linear OLS too simplistic: cannot capture nonlinear relations

OLS Regression Results

Dep. Variable:	CDS_Spread_5Y	R-squared:	0.075
Model:	OLS	Adj. R-squared:	-0.007
Method:	Least Squares	F-statistic:	1.325
Date:	Thu, 11 Sep 2025	Prob (F-statistic):	0.275
Time:	20:23:44	Log-Likelihood:	-216.01
No. Observations:	50	AIC:	442.0
Df Residuals:	45	BIC:	451.6
Df Model:	4		
Covariance Type:	HC0		

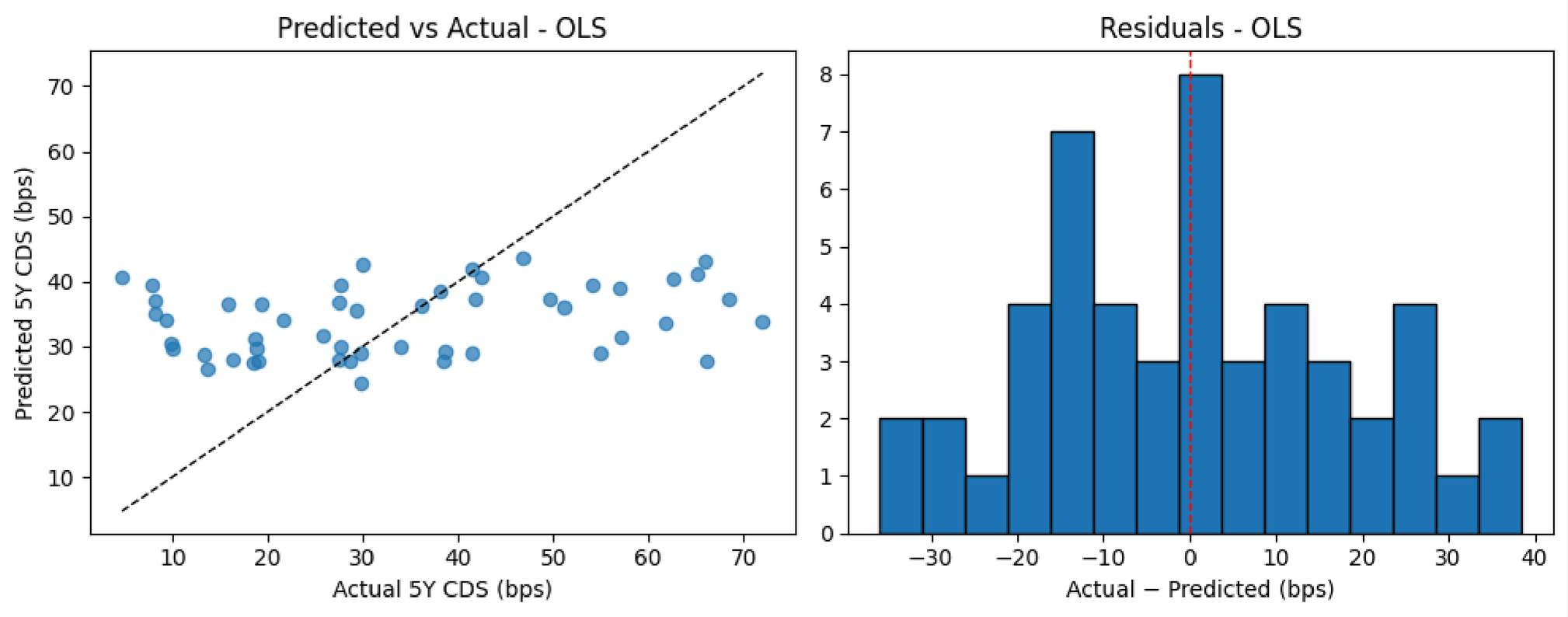
	coef	std err	t	P>|t|	[0.025	0.975]
const	17.8903	14.060	1.272	0.210	-10.429	46.209
ESG_Score	0.0034	0.161	0.021	0.983	-0.320	0.327
Total_Assets	4.114e-05	5.99e-05	0.686	0.496	-7.96e-05	0.000
Leverage_Ratio	5.8857	3.636	1.619	0.112	-1.437	13.209
EBITDA_Margin	0.2503	0.214	1.168	0.249	-0.181	0.682

Omnibus:	1.472	Durbin-Watson:	1.679
Prob(Omnibus):	0.479	Jarque-Bera (JB):	1.280
Skew:	0.226	Prob(JB):	0.527
Kurtosis:	2.359	Cond. No.	5.50e+05

Notes:
[1] Standard Errors are heteroscedasticity robust (HC0)
[2] The condition number is large, 5.5e+05. This might indicate that there are strong multicollinearity or other numerical problems.

- **In-sample** $R^2 = 0.08$
- **Cross-validated** $R^2 = -0.123 \pm 0.072$ (5-fold)

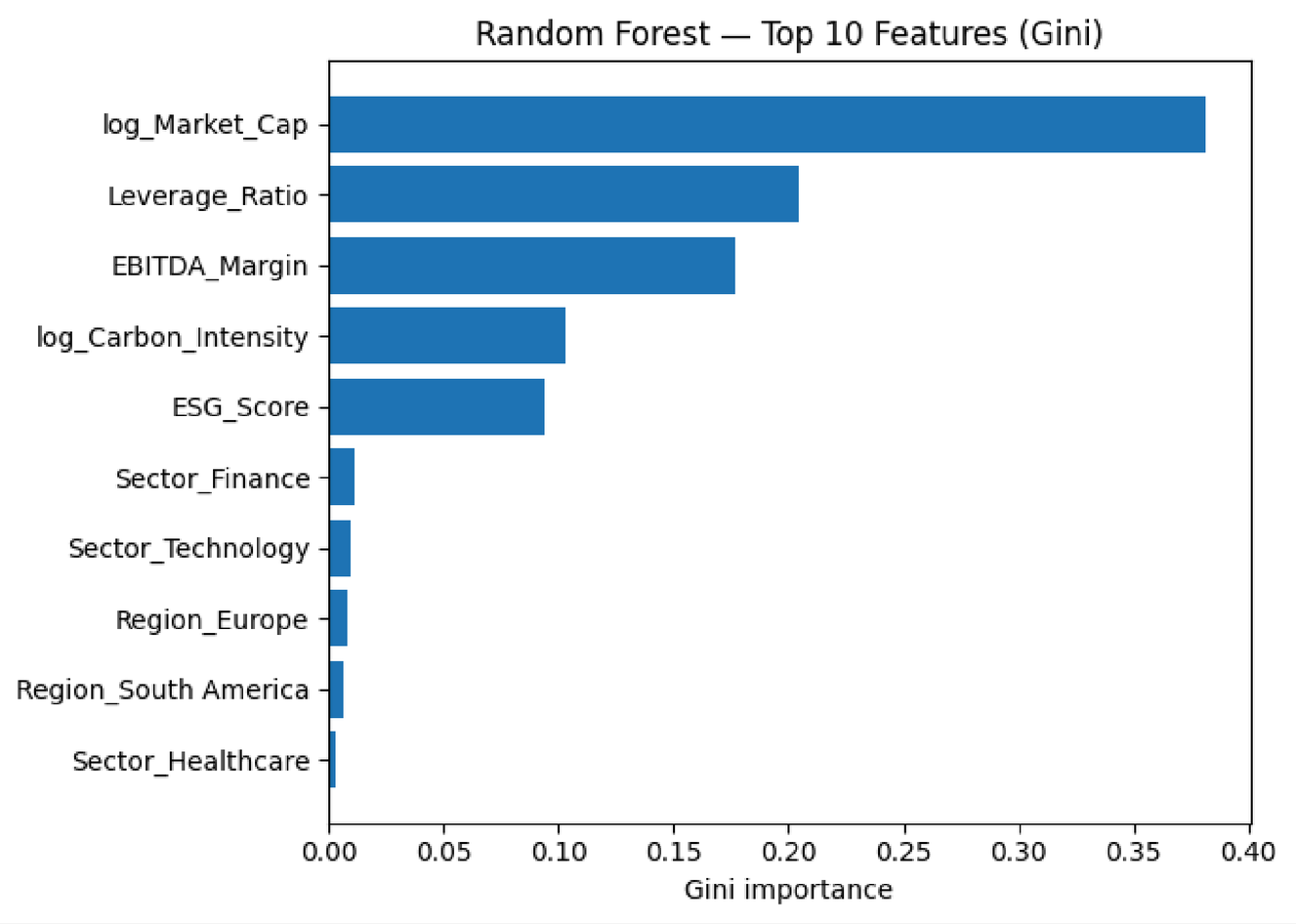
The model fails to capture meaningful relationships



Random Forest

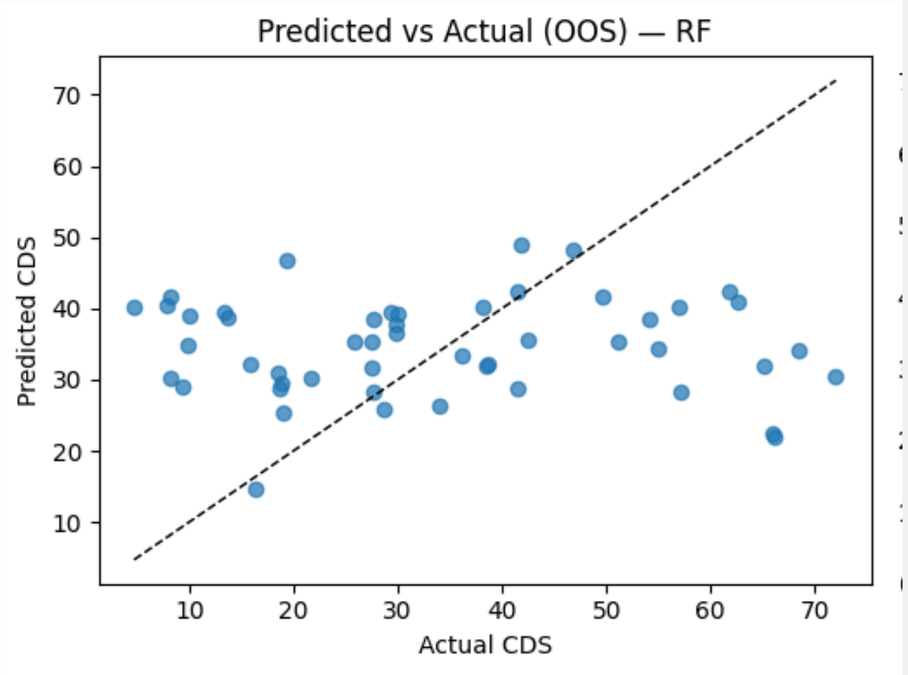
Parameters

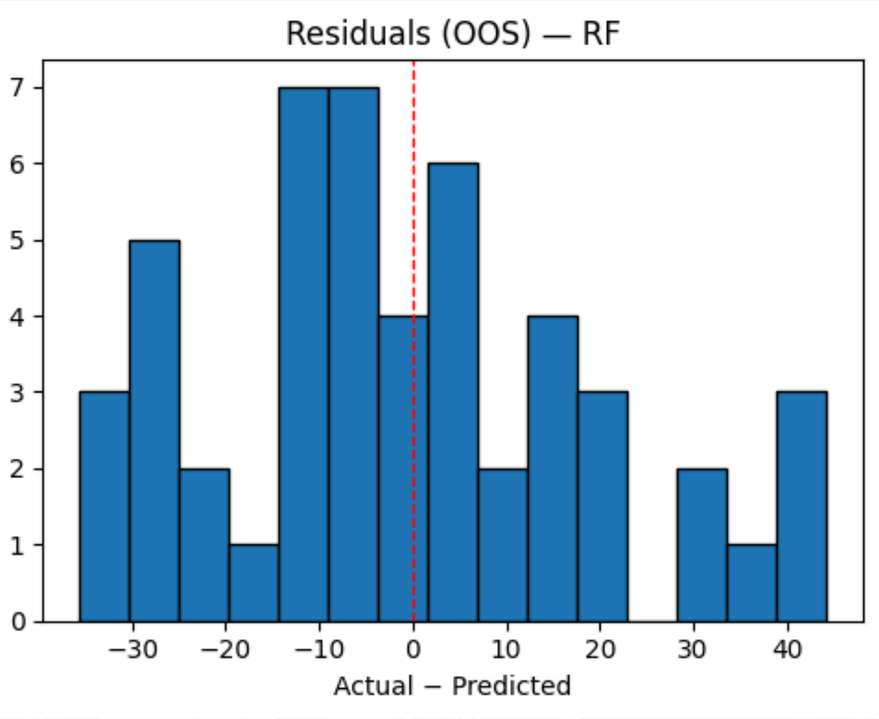
n_estimators=400,
max_depth=3,
random_state=42,
n_jobs=-1,
max_features=0.6,
min_samples_leaf=6



Covariate selection

- ESG_Score
- Log_Market_Cap
- log_Carbon_Intensity
- Leverage_Ratio
- EBITDA_Margin
- Sector (via one-hot encoding)
- Region (via one-hot encoding)





Do not look into statistics!!!!!!!!!!!

- **Cross-validated** performance (5-fold):

  R^2: -0.073 ± 0.161

  RMSE: 19.04 ± 1.14

  MAE: 15.63 ± 0.56

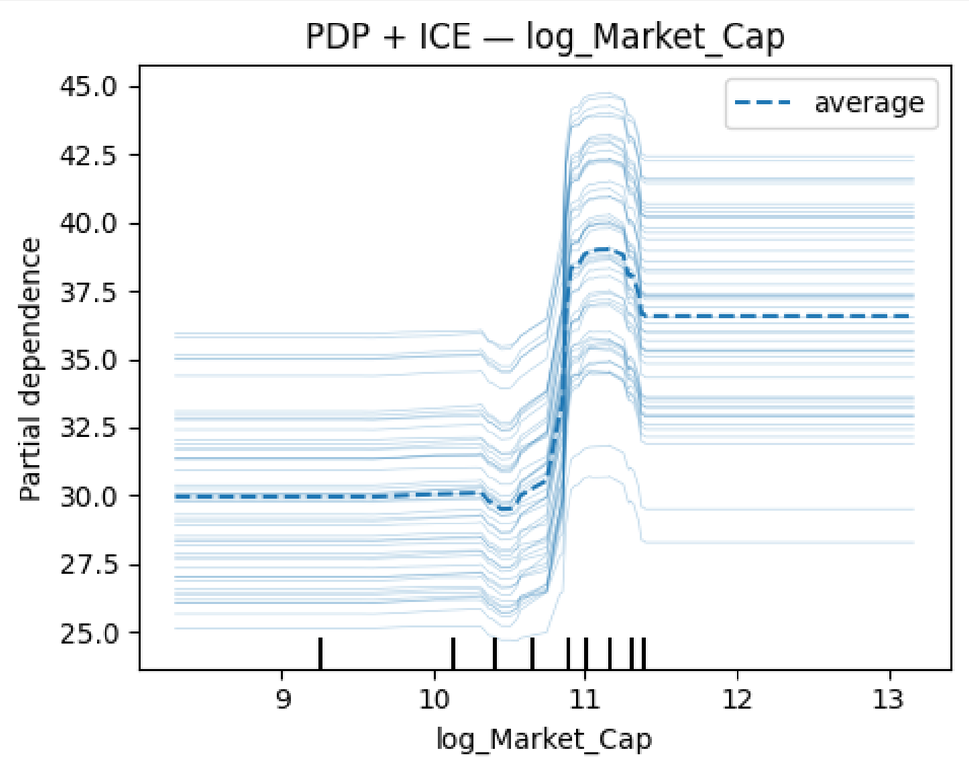
- **Out Of Sample** performance:

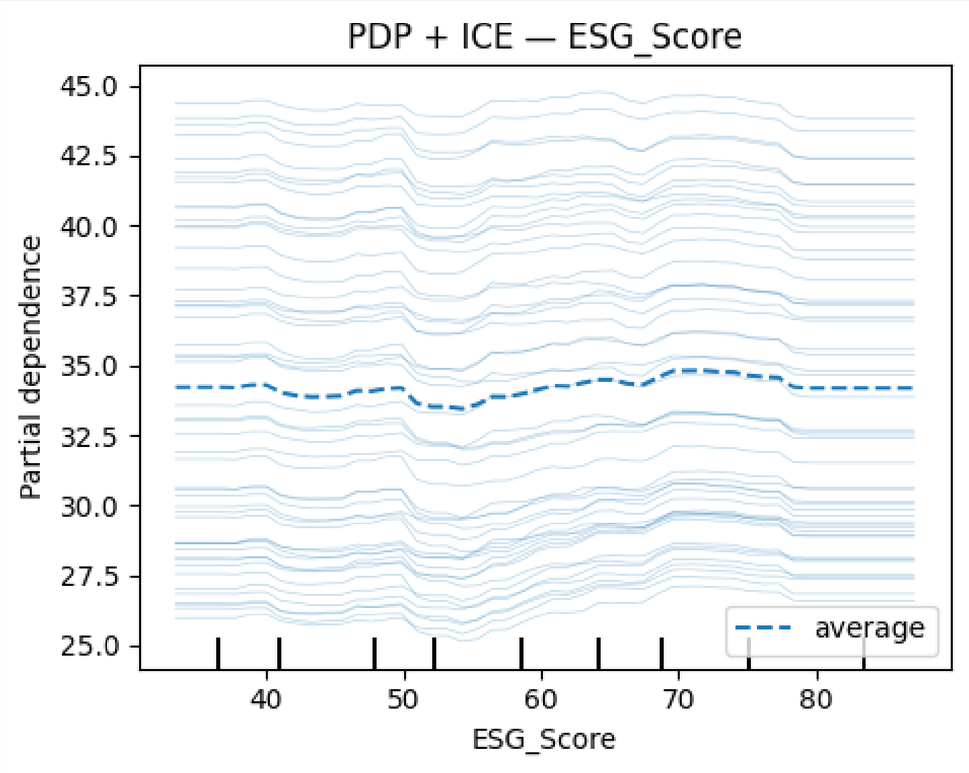
  R^2: -0.017

  RMSE: 19.08

  MAE: 15.63

Partial dependencies comparison - RF





MarketCap:

- Size matters non-linearly.
- Bigger firms → lower predicted CDS around that threshold.
- Thin lines spread out → effect differs across companies (interactions).

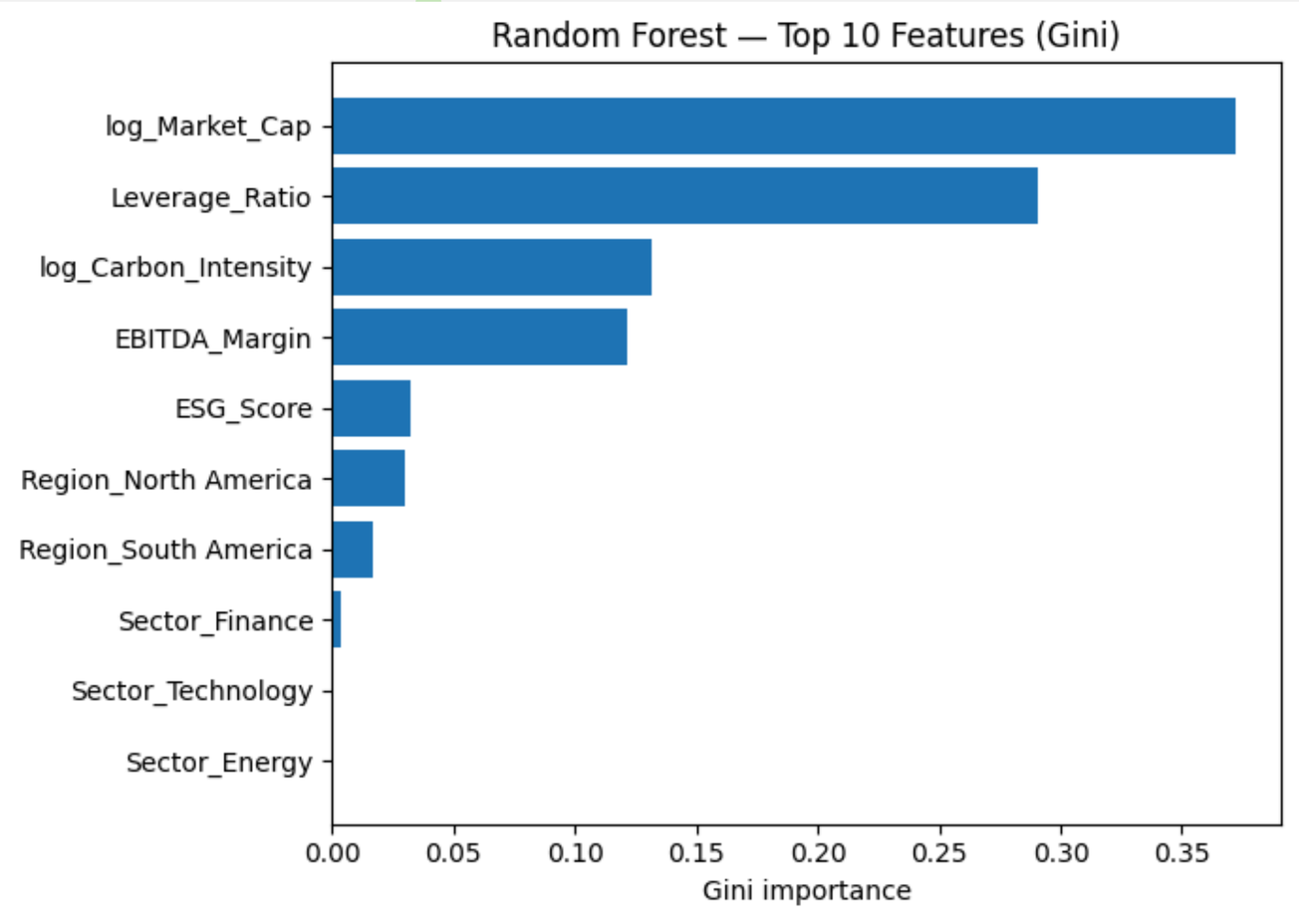
ESG_Score :

- Almost flat average line → very small marginal effect.
- The thin lines are close together → companies behave similarly for ESG.

Gradient Boosting

Parameters

n_estimators=600,
learning_rate=0.01,
max_depth=3,
min_samples_leaf=5,
random_state=42,



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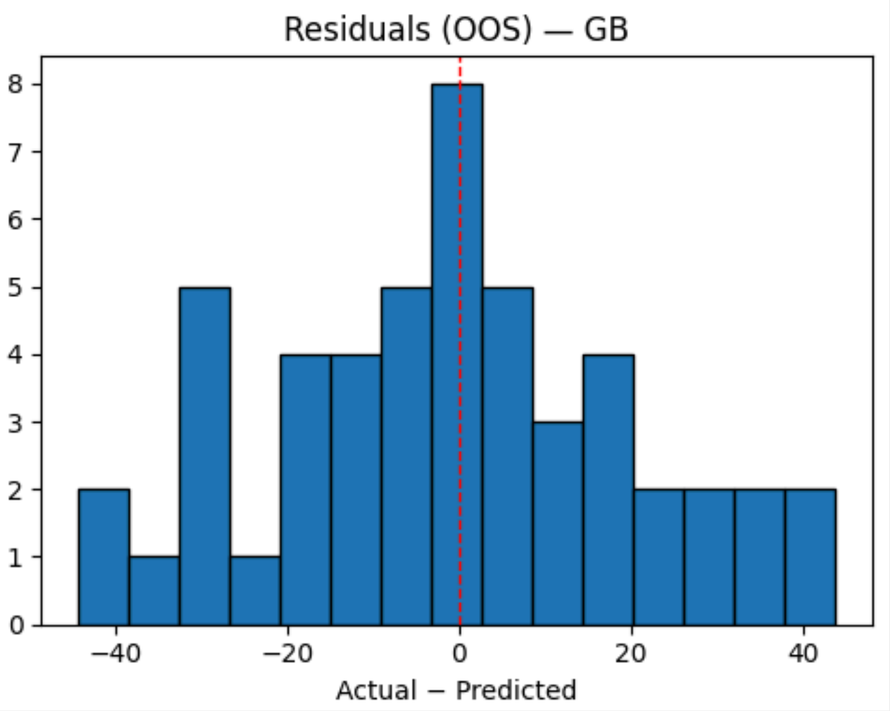


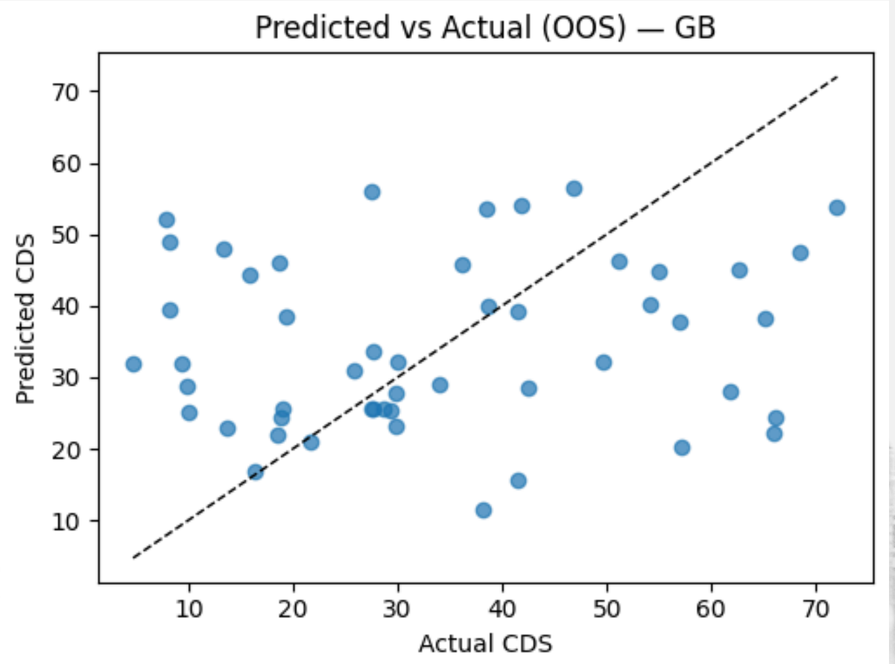
Do not look into statistics!!!!!!!!!!

- **Cross-validated** performance (5-fold):

  R^2: -0.325 ± 0.420

  RMSE: 20.77 ± 2.18

  MAE: 16.49 ± 2.05



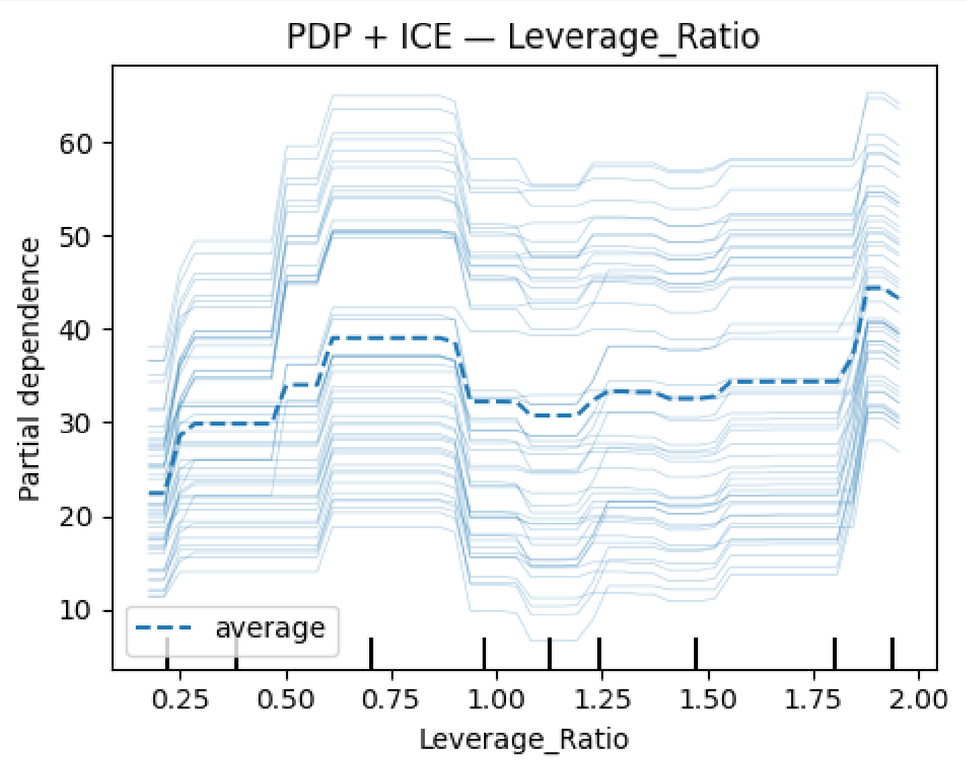
- **Out Of Sample** performance:

  R^2: -0.219

  RMSE: 20.89

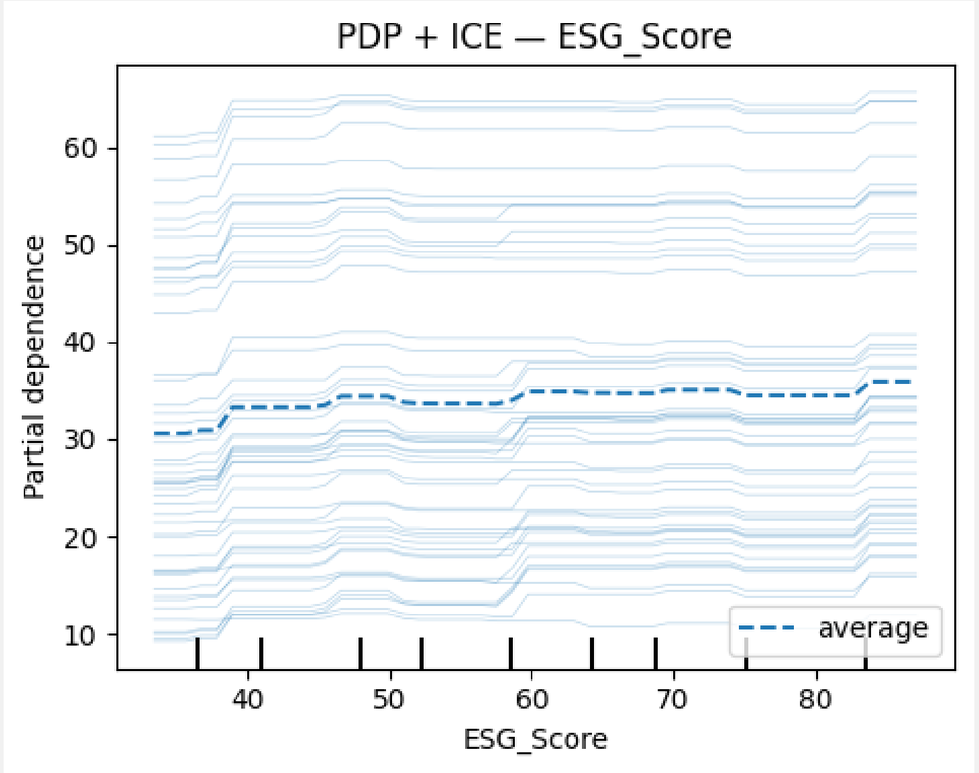
  MAE: 16.49

Partial dependencies comparison - GB



Leverage ratio

- Clear non-linear effect with steps/kinks (≈ 0.4–0.6, ~1.0, ~1.8).
- Overall pattern: higher leverage → higher predicted CDS, with a dip around ~1.0 and a jump near ~1.8.
- Leverage is a strong, non-linear driver of CDS (threshold behavior).



ESG_Score

- Almost flat average line → very small marginal effect.
- Tight ICE bundle ⇒ similar behavior across firms; little interaction picked up by the model.

Non linear model conclusions

- Fundamentals dominate: The model relies on Leverage and Size (Total Assets) first, then EBITDA margin.
- Minor sustainability effect: ESG_Score are secondary—their ranks are only after fundamentals. Carbon Intensity is still bigger factor for CDS.
- Partial dependence = flat for ESG
- Sector/region are minor adjustments after firm fundamentals;
- Gradient Boosting performs worse than a Random forest, due to small dataset
- We can not rely on statistics only!!!

Translate CDS spreads into CVA

Building Blocks

- Piecewise-linear interpolation of the zero rates
- Computation of the exponential discount factor
- Computation of the hazard rate

$$\lambda = \frac{CDS_{Spread}}{1-R}$$

- Computation of the survival probability at each time t_i

$$S_i = \exp\left(-\lambda \cdot t_i\right)$$

- Loss Given Default fixed at 0.6

Final Formula

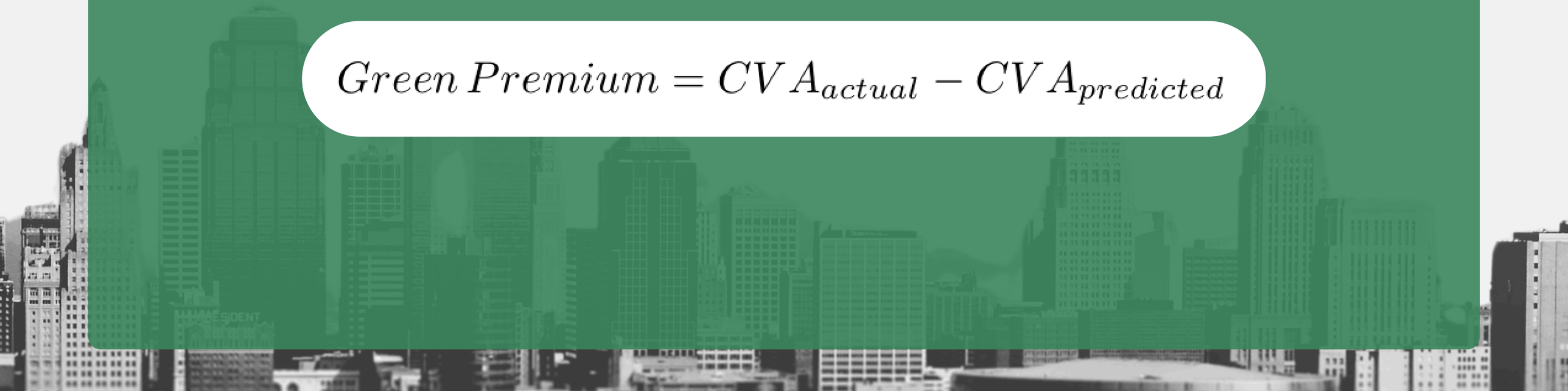
$$CVA = (1 - R) \int_0^T EE(t) \cdot DF(t) \cdot \lambda(t) \cdot S(t) \cdot dt$$

$$CVA \approx LGD \cdot \sum_{i=1}^{N} EE_i \cdot DF_i \cdot (S_{i-1} - S_i)$$

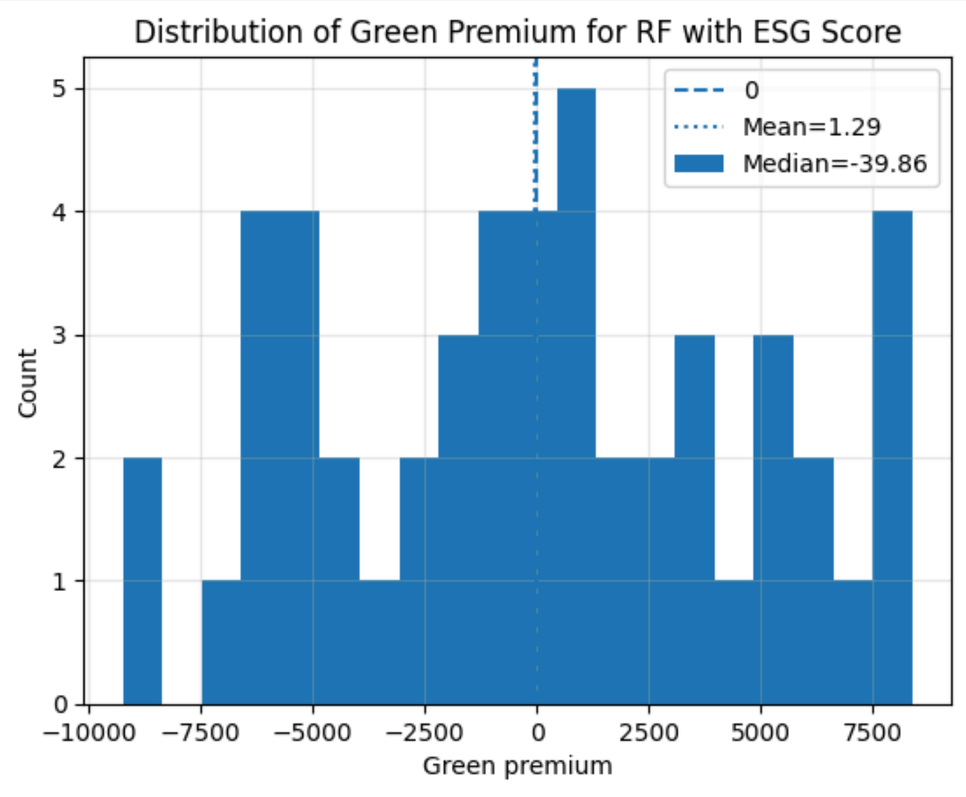
EE_i is given

Can we find a green premium?

$$Green\ Premium = CVA_{actual} - CVA_{predicted}$$



Green Premium - Random Forest



Center ≈ 0 Mean ≈ 1.3 Median ≈ -39.9

No clear skew: both negative and positive tails

Wide spread (≈ −7.5k … +9.5k).

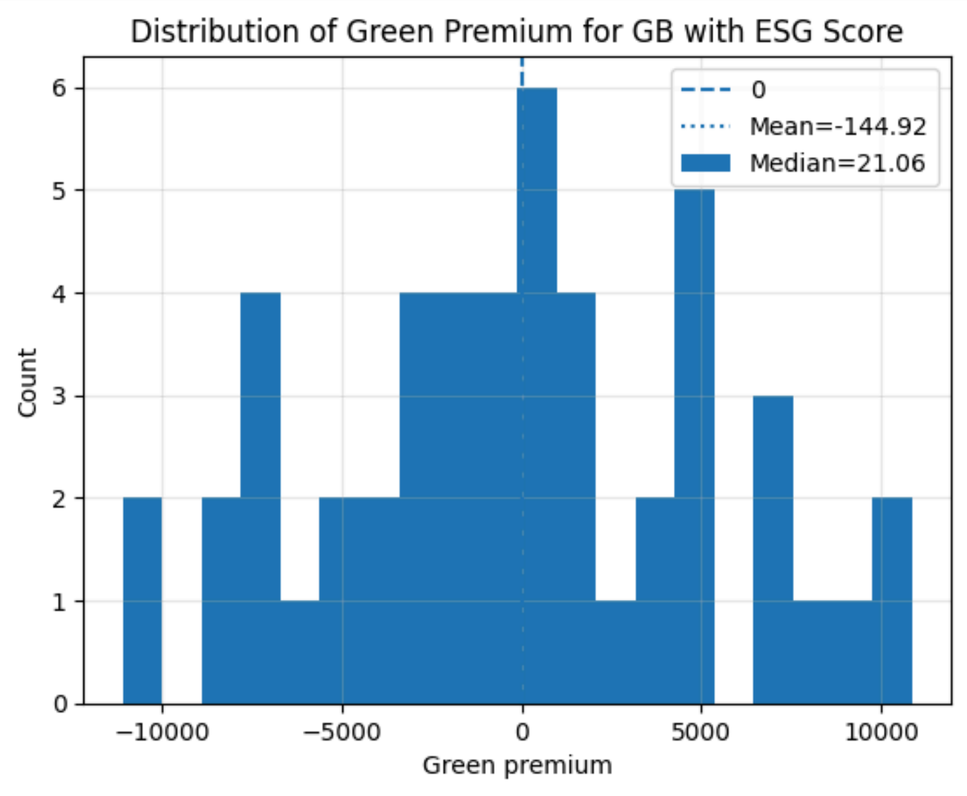
If we divide by the mean of CVA_true we get -1.08 and 0.98 as limits of the x-axis

In relative terms this Green Premium is quite small!

Green Premium - Random Forest

- No systematic green premium. Results are effectively centered on zero
- There is no clear distribution: heterogeneous across firms.
- Consistent with model diagnostics: ESG had weak marginal effect, so ΔCVA concentrates around zero.

Green Premium - Gradient Boosting



Center ≈ 0 Mean ≈ -145 Median ≈ 21

There are both negative and positive tails.

Wide spread. Most firms are clustered near zero, but there is large variation.

If we divide by the mean of CVA_true we get -1.29 and 1.27 as limits of the x-axis

In relative terms this Green Premium is quite small!

Green Premium - gradient boosting

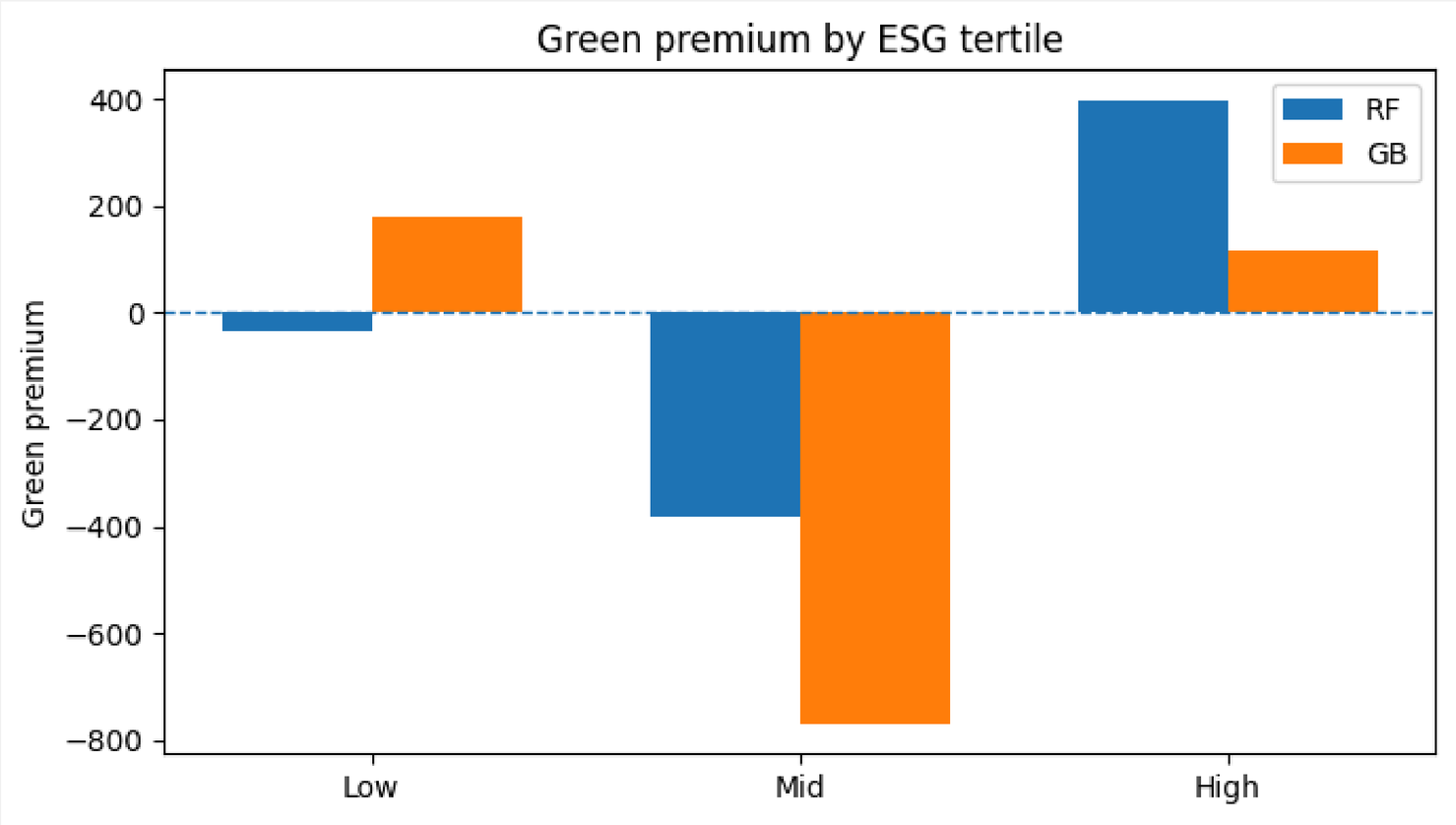
- No systematic green premium. The distribution is roughly balanced around zero.
- Heterogeneous across firms: some names show savings (negative green premium), others show penalties (positive green premium). There’s no uniform effect.
- Consistent with model diagnostics: Corresponds to previous findings from features importance and linear regression.

Division by tertiles

Low ESG: small effect
Mid ESG: clear penalty in both models (stronger for GB)
High ESG: meaningful premium (by random forest)

There’s no uniform green premium across the sample.

Any “green premium” shows only in High-ESG, and it’s not monotonic (Mid shows a penalty).



Final conclusion

The analysis did not yield conclusive evidence of a green premium.

Although the average value appears slightly positive, **data limitations** may have influenced the results, leading to inconclusive findings.

Further research, supported by more comprehensive and high-quality data, is necessary to draw more definitive conclusions.

Thank you very much!